In [2]:
```
import pandas as pd
```

In [3]:
```
import matplotlib.pyplot as plt
```

In [4]:
```
data=pd.read_csv(r"C:\Users\praka\Downloads\Dataset .csv")
```

In [5]:
```
data.head()
```

Out[5]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longituc
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.0275
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.0141
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.0568
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.0564
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.0575

5 rows × 21 columns

In [6]:
```
data.tail()
```

Out[6]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Loc Verb
9546	5915730	Naml\ Gurme	208	––stanbul	Kemanke–– Karamustafa Pa––a Mahallesi, R\ht\m ...	Karak–_y	Karak• ––sta
9547	5908749	Ceviz A––ac\	208	––stanbul	Ko––uyolu Mahallesi, Muhittin ––st–_nda–– Cadd...	Ko––uyolu	Ko––u ––sta
9548	5915807	Huqqa	208	––stanbul	Kuru–_e––me Mahallesi, Muallim Naci Caddesi, N...	Kuru–_e––me	Kuru–_e–– ––sta
9549	5916112	A–––k Kahve	208	––stanbul	Kuru–_e––me Mahallesi, Muallim Naci Caddesi, N...	Kuru–_e––me	Kuru–_e–– ––sta
9550	5927402	Walter's Coffee Roastery	208	––stanbul	Cafea––a Mahallesi, Bademalt\ Sokak, No 21/B, ...	Moda	M ––sta

5 rows × 21 columns

In [7]:
```
data.isnull().sum()
```

Out[7]:
```
Restaurant ID           0
Restaurant Name         0
Country Code            0
City                    0
Address                 0
Locality                0
Locality Verbose        0
Longitude               0
Latitude                0
Cuisines                9
Average Cost for two    0
Currency                0
Has Table booking       0
Has Online delivery     0
Is delivering now       0
Switch to order menu    0
Price range             0
Aggregate rating        0
Rating color            0
Rating text             0
Votes                   0
dtype: int64
```

In [8]:
```
data.duplicated()
```

Out[8]:
```
0       False
1       False
2       False
3       False
4       False
        ...
9546    False
9547    False
9548    False
9549    False
9550    False
Length: 9551, dtype: bool
```

In [9]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
Restaurant ID           9551 non-null int64
Restaurant Name         9551 non-null object
Country Code            9551 non-null int64
City                    9551 non-null object
Address                 9551 non-null object
Locality                9551 non-null object
Locality Verbose        9551 non-null object
Longitude               9551 non-null float64
Latitude                9551 non-null float64
Cuisines                9542 non-null object
Average Cost for two    9551 non-null int64
Currency                9551 non-null object
Has Table booking       9551 non-null object
Has Online delivery     9551 non-null object
Is delivering now       9551 non-null object
Switch to order menu    9551 non-null object
Price range             9551 non-null int64
Aggregate rating        9551 non-null float64
Rating color            9551 non-null object
Rating text             9551 non-null object
Votes                   9551 non-null int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

In [10]:
```
data.shape
```

Out[10]: (9551, 21)

In [11]:
```
data.describe()
```

Out[11]:

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggre ra
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.00
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	2.66
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	1.51
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.00
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	2.50
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	3.20
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	3.70
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	4.90

In [12]:
```
num_cols=data.select_dtypes(include=['int64','float64']).columns
```

In [13]:
```
for col in num_cols:
    print(f"\n Columns:{col}")
    print("mean:",data[col].mean())
    print("median:",data[col].median())
    print("std dev:",data[col].std())
```

```
 Columns:Restaurant ID
mean: 9051128.349178096
median: 6004089.0
std dev: 8791521.282104313

 Columns:Country Code
mean: 18.365616165846507
median: 1.0
std dev: 56.750545600949856

 Columns:Longitude
mean: 64.12657446168703
median: 77.1919642
std dev: 41.467057847617575

 Columns:Latitude
mean: 25.8543807000748
median: 28.57046888
std dev: 11.007935124784685

 Columns:Average Cost for two
mean: 1199.2107632708617
median: 400.0
std dev: 16121.18307349965

 Columns:Price range
mean: 1.804837189823055
median: 2.0
std dev: 0.9056088473975366

 Columns:Aggregate rating
mean: 2.6663700136111426
median: 3.2
std dev: 1.5163775396521686

 Columns:Votes
mean: 156.909747670401
median: 31.0
std dev: 430.1691453762977
```

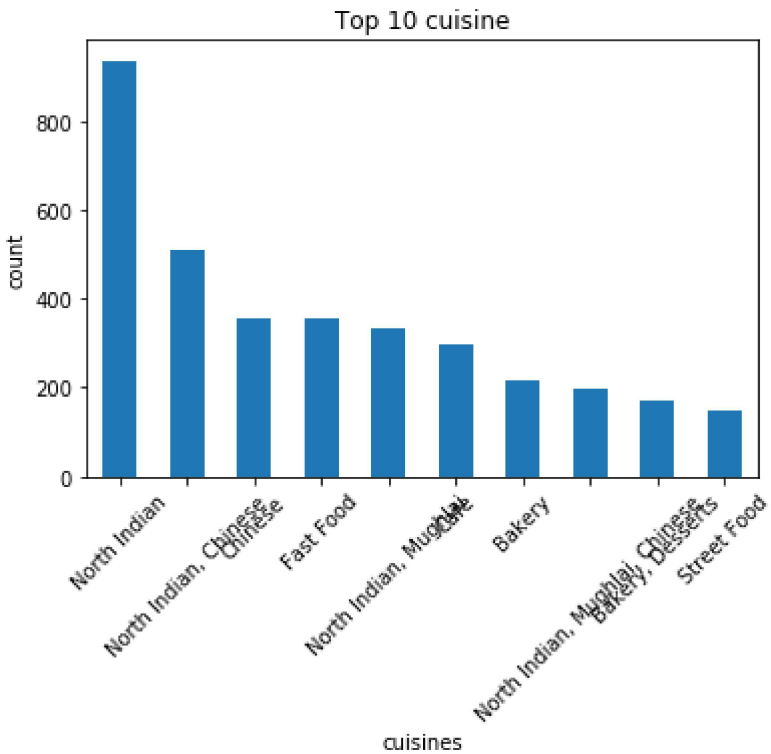
In [14]:
```
print("\n Country Code distribution:\n")
print(data['Country Code'].value_counts())
print("\n city distribution:\n")
print(data['City'].value_counts())
print("\n Cuisine distribution:\n")
print(data['Cuisines'].value_counts().head(10))
```

```
 Country Code distribution:

1      8652
216     434
215      80
214      60
30       60
189      60
148      40
208      34
14       24
162      22
94       21
191      20
166      20
184      20
37        4
Name: Country Code, dtype: int64

 city distribution:

New Delhi         5473
Gurgaon           1118
Noida             1080
Faridabad          251
Ghaziabad           25
                  ...
Bandung              1
Winchester Bay       1
Mayfield             1
East Ballina         1
Weirton              1
Name: City, Length: 141, dtype: int64

 Cuisine distribution:

North Indian                      936
North Indian, Chinese             511
Chinese                           354
Fast Food                         354
North Indian, Mughlai             334
Cafe                              299
Bakery                            218
North Indian, Mughlai, Chinese    197
Bakery, Desserts                  170
Street Food                       149
Name: Cuisines, dtype: int64
```

In [15]:
```
top_cuisines=data['Cuisines'].value_counts().head(10)
print("\n Top cuisines:\n",top_cuisines)
```

```
 Top cuisines:
 North Indian                       936
North Indian, Chinese              511
Chinese                            354
Fast Food                          354
North Indian, Mughlai              334
Cafe                               299
Bakery                             218
North Indian, Mughlai, Chinese     197
Bakery, Desserts                   170
Street Food                        149
Name: Cuisines, dtype: int64
```

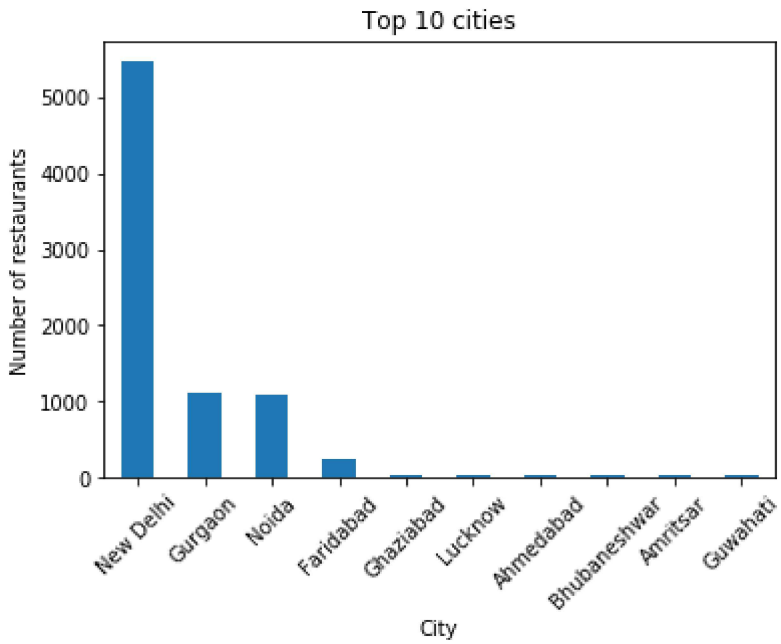
In [18]:
```
top_cuisines.plot(kind='bar')
plt.title("Top 10 cuisine")
plt.xlabel("cuisines")
plt.ylabel('count')
plt.xticks(rotation=45)
plt.show()
```



In [20]:
```
top_cities=data['City'].value_counts().head(10)
print("\n Top 10 cities:\n",top_cities)
```

```
 Top 10 cities:
 New Delhi       5473
Gurgaon         1118
Noida           1080
Faridabad        251
Ghaziabad         25
Lucknow           21
Ahmedabad         21
Bhubaneshwar      21
Amritsar          21
Guwahati          21
Name: City, dtype: int64
```

In [21]:
```
top_cities.plot(kind='bar')
plt.title("Top 10 cities")
plt.xlabel("City")
plt.ylabel('Number of restaurants')
plt.xticks(rotation=45)
plt.show()
```



In [ ]: